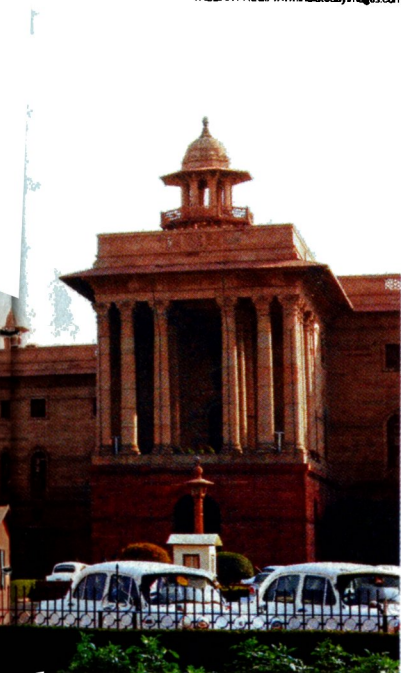




Cyber War Threats

Intelligence Gathering: By snooping in on computers by planting viruses in them.

Information Warfare: Acts to gain superiority by affecting the adversary's information-based systems and computer-based networks while defending one's own networks.

Insider Threat: The most difficult aspect of cyber warfare and security. Insiders have knowledge of your vulnerabilities and configurations. It is easier for them to launch attacks and hide behind anonymous sources.

Hacking: More of a nuisance than strategic value unless people are careless enough to put sensitive information on their websites.

Cyber Crime: Web-based criminal attacks.

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SANJAY PANDEY, CEO, iSec Services



network. It was traced down to commercial Internet accounts located in Hainan island, China. GhostNet, the investigation revealed, comprised at least 1,295 infected computers in 103 countries. Close to 30 per cent of the infected computers were high-value and included those in foreign ministries of India, Bangladesh and Iran. The cyber-espionage ring also targeted computers used by the Dalai Lama and the Tibetan government in exile. The GhostNet directs infected computers to download a Trojan known as gh0st RAT that allows attackers to gain complete, real-time control of the targeted computer, including searching and downloading specific files, and covertly operating attached microphones and web cameras to snoop on meetings. “GhostNet spreads through social means. Contextually relevant e-mails are sent to specific targets with attached doc-

uments that are packed with exploit code and Trojan,” the study says. This modus operandi was repeated in the attack on Indian Government computers in December. A dormant .nic.in address provided the Trojan its cover. A 2007 Intelligence Bureau circular had warned government officials against using personal e-mail ids like gmail and to stick to the .nic.in addresses, which evidently, were thought to be more reliable.

While the cyber espionage attacks on the South Block computers were reported, experts say there are thousands of “mapping attempts” that go unreported. For several months, they have detected Chinese hackers “scanning ports” in Indian networks or “laying the electronic battlefield”. “This is the equivalent of a burglar testing out the doors and windows of your house,” says a senior intelligence official. In two years the intelligence agencies

have built a database of over 20,000 such Chinese IP addresses from where attacks have originated.

With cyber attacks now resembling political conflict, experts warn of the unfolding of a “Cyber Cold War”. “In all probability, the next war is going to be preceded by a massive cyber attack. Peacetime probing attacks are being carried out against China’s imagined, existing and potential adversaries,” says security analyst RSN Singh. India is extremely vulnerable. It has seen a huge growth in state wide area networks and 25 mission mode projects, besides electrical grid, oil and gas infrastructure. This is in addition to the private industry, especially the banking and finance sector where huge number of transactions are conducted online. The Chinese have developed a secure operating system “Kylin” to protect themselves from attacks. They have also developed a secure microprocessor that, unlike US-made chips, is known to be hardened against external access. “The Chinese can keep attacking us, but we can never attack them,” says a senior intelligence official.

India, despite its status as an IT-superpower, is almost entirely dependent on external sources for hardware and software and, most importantly, all its anti-virus, network protocols and network hardware components. These vulnerable operating systems could be used for “distributed denial of service” or massive DDoS attacks, of the kind used by Russian hackers to paralyse Georgian servers during the South Ossetia war in August 2008. Experts suggest India should begin writing an operating system from a scratch. There is a need to increase the cyber warfare budgets and induct more cyber warriors, particularly those well-versed in Chinese. “We need a dedicated cyber security organisation staffed with police and armed forces personnel to start developing a strategy to protect national information infrastructure,” says Sanjay Pandey, former IPS officer and CEO of Mumbai-based iSec Services. Clearly, the need of the hour, if India is to protect itself from the snooping dragon. ■